WOWII: A counterexample to conjecture 155

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1 Conjecture

If G is a simple connected graph, then

$$L_s(G) \geq 1 + \textit{number of distinct orders of radial circles}$$

Conjecture reference.

Notations

$L_s(G)$: **maximum number of leaves of a spanning tree**. A spanning tree of G is a subgraph that contains all the vertices of G and is a tree. The number of leaves of a tree is the number of vertices of degree 1 in this tree. Thus, $L_s(G)$ is the maximum, among all spanning trees of G, of the number of vertices of degree 1.

Radial circle: a radial circle is the set of vertices located at a given distance (radius) from a center vertex. For a center vertex v and a radius $d \geq 1$, the radial circle is the set of vertices u such that $\text{dist}(v, u) = d$.

Order of a radial circle: the order of a radial circle is the number of vertices it contains, that is to say the cardinality of this set.

Number of distinct orders of radial circles (k): we consider all radial circles for all choices of the center vertex v and for all radii $d \geq 1$ such that the radial circle is non-empty. The parameter k is the number of distinct values obtained among the orders of these radial circles.

2 Results

2.1 Graph identification

The graph is defined by the following graph6 string:

```
WhCGGC@?G?_@?@??_?G?@??C??G??G??C??@???G????G?@
```

2.2 Calculated properties

Property	**Value**
Number of vertices n	24
Number of edges m	23
Max number of leaves $L_s(G)$	3
Distinct orders of circles k	3
Set of orders	$\{1, 2, 3\}$

Table 1: Properties of the counterexample

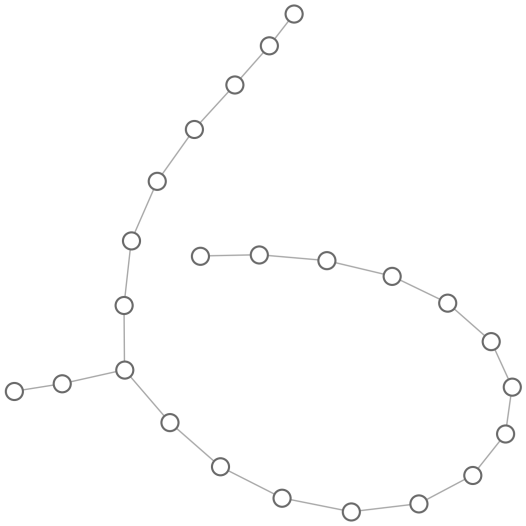


Figure 1: Counterexample

3 Verification of the inequality

1. Calculation of the left-hand side (LHS): Since the graph is a tree with only 3 vertices of degree 1, any spanning tree is the graph itself.

$$LHS = L_s(G) = 3$$

2. Calculation of the right-hand side (RHS): The number of distinct orders found is $k = 3$ (the possible sizes of the sets of vertices at distance d are 1, 2, or 3).

$$RHS = 1 + 3 = 4$$

4 Conclusion

The conjecture states that $L_s(G) \geq 1 + k$ where k is the number of distinct orders of radial circles. However, we observe:

$$L_s(G) = 3 \quad \text{and} \quad \text{RHS} = 4$$

$$3 < 4$$

The inequality is not satisfied. **This graph is therefore a valid counterexample.**